

SECTION 1 — INTRODUCTION & OPENING REMARKS

MODERATOR / CHAIR (American Kratom Association)

“Could we get everybody to take their seats and we will start this hearing. Sounds great, that’s a good... that would do it. Good morning, and thank you all for being here today for this important Congressional briefing on kratom.

I want to extend my sincere gratitude to Representatives Mark Pocan of Wisconsin and Jack Bergman of Michigan for their leadership as co-sponsors of the federal kratom Consumer Protection Act, or KCPA. Their tireless work has been pivotal in ensuring that the voices of millions of American consumers who rely on kratom for their health and well-being are heard.

We’re also fortunate to have members of your staff from both of the congressmen, and they specialize in FDA issues, with us today. I’d like to personally acknowledge their presence and thank them for their efforts to ensure evidence-based policy protects the rights of Americans to make informed choices on how kratom can help Americans with making informed choices to support their health and well-being.

We understand that potential votes and other scheduling demands will require some participants to come and go as needed, but the briefing is being recorded and will be available for review in its entirety.

As chairman of the American Kratom Association and a former member of Congress myself, I understand firsthand the importance of providing members and their staff with accurate, science-based information to help guide their decision-making process.

The purpose of today’s briefing is to share with you the latest advancements in scientific research and public policy that shed light on the safety profile of kratom and real-life experiences of consumers across the nation.

We have an outstanding lineup of experts here today who will provide invaluable insights. Our first panel will feature a group of leading scientists who have dedicated their careers to studying katum and its impact on health, and I'm honored to introduce them. And I will then introduce the second panel, and we'll... we'll get off to the races.

First of all, we have Dr. Christopher Mccy from the University of Florida, Dr. Edward Boyer from Ohio State University and Harvard Medical School, Dr. Kirston Smith from John's Hopkins University, and Dr. Jack Ja Henningfield to my left — is that appropriate today? Okay, on the political spectrum I don't... I don't know. Dr. Jack, Jack Henningfield, also from Johns Hopkins.

These distinguished experts will share their groundbreaking research on katum, offering evidence-based perspectives on its safety and efficacy.

Dr. Nora Vulco, Director of the National Institute on Drug Abuse, Naida, we deeply appreciate her continued support researching the potential benefits and risks of katum use. We're grateful for her thoughtful contributions to today's discussion.

Following our scientific panel, we'll hear from passionate katum advocates who've witnessed firsthand the positive impact katum can have on individuals and communities. Their personal stories and advocacy efforts will reflect the very real positive impact katum has had on their lives and countless others. I will introduce them once the first panel concludes.

As we move forward today, I encourage you to listen with an open mind and consider the significant role that science, experience, and common-sense regulation can play in ensuring that katum remains a safe option for consumers who rely on it for their well-being.

Once again, thank you to everyone who's made this briefing possible. Let's work together to protect katum consumers and ensure that sound evidence-based policies guide the future of katum regulation in the United States."

MICAH BARROW (Staff for Rep. Mark Pocan)

“[Applause] Yes, so thank you. Can you all hear me okay? Thank you all so much.

My name is M. Barrow. I help cover health-care issues for the Congressman. And again, this is a brief statement from him. My apologies that you are with me today instead of him. He is very sorry that he wasn't able to make it, but he got called back to the district at the last minute yesterday.

So thank you all for being here. My apologies for him being unable to be here today with everyone. You are in for a fantastic panel of experts and individuals with lived experience, especially Miss Sakur from my home state of Wisconsin.

As a member of Congress, I have worked with federal representatives in both parties to continue the research and legal use of kratom, due to its promising help in a number of health conditions, as well as its ability to help many people overcome addiction. I've been moved by the many personal stories of people across the nation who use kratom. The data and stories that you hear today will be able to highlight the importance of—”

REP. JACK BERGMAN (R-MI)

“—legislation like my bill, the Federal Clarity for Kratom Consumers Act, which would require the FDA to hold a hearing and establish a task force on the health and safety of kratom products and provide consumers with some more clarity on how the FDA regulates kratom. I'm grateful to everyone here today for engaging in this important conversation, and again, my sincere apologies for being unable to join you.

Thank you all. Thank you, Micah.

Congressman Bergman — that's me. Thank you, by the way. Should I sit like this so you can just see me like that — is that okay? You spend half your time adjusting your positions around here; in this case it was just adjusting the chair you're sitting in.

Here is the Veterans Affairs Committee hearing room that was just, if you will, redone as part of the Cannon renovation here, and it's probably, if not the most impactful, elegant, powerful design of a committee hearing room that is anywhere here on Capitol Hill, because it represents — not in some cases the soul of America, it represents the individual souls, the men and women who have served our country to ensure that we are still here.

So normally, when we're having a committee hearing, I sit up there on that dais. But we have to remember that no matter where we are in the seniority list, we all start out as that brand-new private or that brand-new second lieutenant, for those of us who served, or that brand-new employee of a company, or that brand-new entrepreneur, or that brand-new something. We all started at the beginning; we don't start somewhere in the middle.

But we get to where we're going because we keep doing the right thing for the right reason.

Having said that, I'm sorry that Mark couldn't be here today, because he's a great, great partner in this. The only thing I feel bad for the American people for is that you do not see America — Democrats and Republicans — working together all day, every day, across the aisle to do good policy and good law.

And we know we've got challenges. We know we've got a nationwide mental-health crisis. We know that there are issues out there that need to be solved.

And let's get specific here with katum. It has significant potential to help overcome addiction, to treat withdrawal, to basically get people back on the road to a successful life.

Now Luuk and I — from Los Angeles area — we came in together as freshmen. Luuk is a Democrat, and we are partners in the PATH Caucus. For those of you who may or may not know, PATH stands for Psychedelic Advancing Therapies — new ways to look at things for better mental-health outcomes.

We believed also that really the name should be “GOOD PATH,” because who doesn’t want to be on a good path in life, right? But in this case, GOOD would stand for Get Off Opioid Dependency.

So when you think about, in the katum space and what you’re doing and have done — how do we get people off dependency to other things, especially opioids? That is the challenge.

But we can’t have it be the Wild West. There’s got to be significant involvement of the medical-research community, of those who would create the products, the community, the folks like us as members of Congress to get behind good things and to ensure that — and it’s largely policy more than law — we have good solutions put in place.

So, when we — meaning Mark and I — introduced H.R. 5905, which is the Federal Katum Consumer Protection Act... we have long names for all these bills, and we usually give them an acronym, and then nobody knows what the acronym really stands for because they all sound alike. But the point is: you don’t... you don’t dictate; you work with others. You work with the FDA, you work with industry, you work with therapists, you work with the academic researchers. This is all about with and together.

Having said that, I’m looking forward to hearing the panels today, because just know you have advocates here in the elected space that are with you. I know you’re here because you have an interest in the space; you’re already involved in better outcomes and better mental health, which will hopefully lead to better physical health for people around the world.”

DR. CHRISTOPHER MCCURDY

“Thank you very much. I’m Chris McCurdy.

We were joking last night a little bit that, if you go back to ground zero of research and science in the United States, I think I might be it. We started working on this plant almost 20

years ago, long before it was even recognized by the DEA or the FDA, because we saw potential all the way back then.

And so I want to bring you up to date from where we've come in those 20 years along a tremendous journey — where we didn't know much — and in the last five years we've been part of over 60 million dollars' worth of federal funding to investigate this plant and its medical potential.

About 12 million dollars of that came to the University of Florida directly under grants that I supervised, along with our team, and the other 50 million was at the National Center for Advancing Translational Sciences — or NCATS — which is moving toward the first clinical trial of mitragynine. That grant provided five kilograms of purified material in pharmaceutical grade that will go into clinical trials, hopefully in the next year.

So we are looking at this from not only a botanical standpoint and the promise of the plant, but indeed we want to see if there is a true pharmaceutical path that could be pursued.

I'm not here to get rid of the plant from the commerce perspective. I want it to be accessible to everybody — always do, always have. Never has it been my intention to turn it into only a pharmaceutical product. However, just like we've seen with other plants and other compounds, we know that some of these compounds can be isolated and beneficial at a highly controlled clinical level as a pharmaceutical.

So that's to just say what we've been doing and where we're headed.

The pictures that you can see on the slide above you are actually kratom trees growing in Malaysia on the very top, and at the bottom are kratom trees growing in Apopka, Florida, at the University of Florida.

We have been able to replicate a lot of the work — a lot of the understanding of what's happening in Southeast Asia — for scientific purposes at the University of Florida, and we

have probably the most soup-to-nuts program in the world investigating from seed all the way to consumer what is happening. And I'm very proud of the support that we've had — of advocates, the industry, the federal government, and the State of Florida — to get to this point.

Okay, so the obvious question that everybody first asks is: is kratom a threat or a therapeutic?

Of course, everyone in this room knows, but I'm going to go backwards a little bit and make sure that on the record we have a good understanding of what the foundational start is.

So — is it a threat or is it a therapeutic? We know it's been anecdotally used for centuries for chronic pain, for mood elevation, for energy, for opioid-use disorder, and alcohol-use disorder now in the United States more recently. We know that it has potential in stimulant-use disorder in lessening the desire for methamphetamine.

There are adulterated and contaminated products that have been reported, and that has put a black mark on the whole industry as a whole. There are claims that kratom is severely addictive and deadly, and our job as scientists is to figure out where all of these claims fall.

So we know that there's potential benefit from a medical standpoint, but we know there's potential harm, and understanding where that line is is vitally important in making sure that this product is used properly and safely.

I've been able to consult with the U.S. Food and Drug Administration most recently on their single-ascending-dose trial — which I'm not going to speak about today, but I'm happy to answer questions on. I've spoken to the World Health Organization and the United Nations in briefings on this as well. And all of those agencies had looked at one time or another to ban kratom from either the United States or globally.

None of that has happened — and a good reason that none of that has happened is most of the people that are sitting at this table right here, the research that has been conducted that has shown some efficacy.

We can't say that it's entirely safe, and we can't say that it's entirely efficacious yet, but we're certainly getting closer to being able to make those statements with scientific validity.

I don't do very much of this work — I direct a huge team — and I acknowledge them every time before I really get going, because if it isn't for those folks, I wouldn't be able to talk to you and share all the work that we've done.

I'm not going to go through and name everyone... I would like to point out Scott A. here in the room — he's been very important in helping us put those trees in Apopka and really getting our program off the ground from a horticultural standpoint.

Kelly Dunn is another person that really helped us from the very early days — he funded our research before anyone else would and really got us enough preliminary data to get things going.

And finally, Jeremy Steing has been incredibly important in the whole process — as a UF alum and as a supporter of our work as well. So thank you to all of you, and thank you to the National Institute on Drug Abuse and the taxpayers for funding so much of this work.

So — we all know *Mitragyna speciosa*. Something that people may not know is that the whole genus *Mitragyna* was named after the fact that the leaf is shaped like a miter hat — or a bishop's hat. If you take that leaf and turn it 90 degrees counterclockwise, it is supposedly shaped like a bishop's hat — and that's where the genus got its name.

Mitragyna speciosa is one of 10 species in that genus. It's a member of the Rubiaceae family — the coffee family — and it's found in Southeast Asia.

But what I do want to point out is something that is vitally important — and that's kratom use in Southeast Asia.

The traditional use of this plant — and these pictures are pictures I took when I was in Malaysia — of a tree in the wild. And then up in the top right is: once those leaves are plucked from the tree, they're quickly boiled into a decoction — it's not really a tea — that boiling takes place for 2 to 3 hours to try to extract as much of the alkaloids out of that. And then those alkaloids are packaged up in baggies or in plastic bottles that are sort of takeaway from the roadside stand that you see there in the middle, where folks can take those with them for their daily consumption.

So it's not too different — if you think about it — from a coffee shop in the United States where someone will go and get their coffee for the morning.

Many people stop by these roadside stands — which are completely criminalized and illegal in Malaysia — but they're there, and most people are dealing with that every day.

So the tea is really utilized by these field workers for pain relief, for stimulant effects to improve their work capacity, to reduce their opioid withdrawal — because many of these rural farmers are actually using raw opium as their pain treatment. And when they run out of access to raw opium, they actually increase their utilization of kratom juice and they're able to avoid withdrawals.

And that was really what pointed us in the direction of saying: hey, there might be something from nature that can help us in the opioid crisis. And opioids are from nature. And I've always believed that there's a balance in nature — that if there's a problem, there's a solution. We just have to be able to recognize where that is and find that.

Leaves are chewed in Thailand — so it's a very different method of administration in Thailand.

And then most kratom raw material in the USA, of course, is from Indonesia.

So even though these are all in Southeast Asia, you're still talking about differences in geographical regions — almost if we were to talk about wine regions of the world — terroir makes a difference. And we've studied — we've been fortunate enough to study — plants from each of these areas.

And one thing that's good is that the alkaloid consistency is there in all regions, including Florida and other commercial ventures in Hawaii.

We know that the predominant active agent — that we believe is the predominant active agent — is mitragynine, a corynanthe-type alkaloid.

And this is the picture of kratom use in the USA — which we know is very different.

So we go from fresh leaf being consumed directly as it's picked off the tree to dried leaves. And you don't have to think too long or hard about it — dried spices are different than fresh spices, and so you can certainly realize there is a difference.

Current estimate in the USA is more than 15 million users. I think this was a number that came out in 2019 pre-pandemic. I think we're well over 20 million on a conservative basis. And I say that because every day I drive home I pass four gas stations — and they always had CBD signs in them — but in the last couple years they all have kratom signs. So I know it's become more mainstream.

One of the things that we've tried to make a strong point about as a team of scientists is that not all kratom is equal — and there's an important distinction between native leaf and extract products. I like to look at it as: we consider everything kratom, just like we consider the entire spirits industry “alcohol.” So you have products like seltzers or light beers, and you have products that are 190-proof Everclear. And now we're in that sort of realm in the

kratom space. But we call everything kratom — we don't differentiate between those. And there's a significant difference.

There's a lot of alkaloids that we've isolated over time, and we're studying those alkaloids.

Therapeutic potential — what we're really interested in — is opioid detoxification. We're looking at medication-assisted therapy — very much like buprenorphine or methadone — is this something that could be utilized to help individuals transition off hardcore opioid prescription or illicit use to something that we consider to be much more effective... although I can't say that officially yet.

Lack of real standardized products has made clinical trials very difficult.

To try and get around that end — this was our effort — as you already saw: the University of Florida growing trees. That's Brian Pearson, who most recently left our program and went to Oregon to head up a research station in Bend, Oregon, which kind of was a dream job for him — to sit on the river and in the mountains and enjoy cherries instead of kratom.

We have worked with our plant counterparts and we have figured out many of the biosynthetic pathways within the plant. We have worked to develop the profiling of the leaves — and this is important.

I had a student from Thailand working in our lab, and she taught us how to chew kratom leaf. And she made a very good point that I'd never realized before in all the time that I've been over to Southeast Asia — but they specifically go to the third or fourth leaf bunch on the branch. Leaves that are fully mature, they will chew. They won't chew the newly emerged leaves.

So I said: there's something scientifically there.

And what we found — and it's hard to see in this slide — but what we found is once you get into those more mature leaves, the mitragynine concentration is maximum.

In the early emerging leaves, another alkaloid species, speciogynine, is the most abundant. And why I say this is important today is because what we've seen in the marketplace is a shift in alkaloid content.

Mitragynine is still the first and most abundant alkaloid. Speciogynine used to be one of the least abundant, but now we see it as the second most abundant alkaloid.

What we think is happening, hypothetically, based on the science we've done, is that the demand is so great that the suppliers can't keep up with the demand — and so they're harvesting less mature leaf material. That's creating this change in alkaloid profiling.

Why is that important?

This is a snapshot of many commercial products that we've done in collaboration with Dr. Smith — looking at a wide swath of the United States marketplace. We haven't published this yet — we're up to about 360-ish samples from all over the United States.

What we found was that it's very consistent in the marketplace — the alkaloid profiling is very consistent. Mitragynine is the major thing.

And importantly — 7-hydroxymitragynine is almost non-existent now in the marketplace in natural leaf material — and I think a lot of our work helped to drive that desire to eliminate 7-hydroxy.

However — as many of us know — the market has seen an infiltration of 7-hydroxymitragynine isolate or synthetic 7-hydroxymitragynine components.

This is the first time this is publicly being shown anywhere.

On the left — that's a fingerprint of what the alkaloid profiling looks like in native leaf. Most products we've tested, including extracts, meet this fingerprint. That big blue arrow on the left is where 7-hydroxymitragynine would be — and it is NOT present in leaf materials.

If we look at the graph on the right — that is a mass-spec tracing of a 7-hydroxymitragynine tablet or product that's in the marketplace. And you can see it's almost the most abundant thing there. But what is the most abundant thing is mitragynine, and this is synthetically created from a mitragynine isolate or concentrate. And then there's several compounds in that tablet that we have no idea what they are — meaning we have no idea what the harm could be or the risk, or what's there.

Anybody — including someone who watches Sesame Street — “which one is different and which one is similar” — you can definitely see that the one on the right is totally different from the one on the left.

And this is concerning for all of us as scientists because this product is making a huge impact in the marketplace right now — and it is NOT kratom. The bottom line is: this is being marketed as a kratom product — and as I said earlier, we lump all things together as being a product — and this is something that needs attention of authorities.

We've alerted many of them already — and we're working on putting out more information on what's going on with this.

The last thing I'll say is that what's frightening from this perspective is: no forensic lab in the United States has the capability to search for 7-hydroxymitragynine. They all use mitragynine as the surrogate for kratom use. And if someone was to have an adverse event or overdose on this product, it will show it being kratom — because it only analyzes for mitragynine.

So we're working with forensics trying to change the way forensic labs in the United States even look at these products — to increase the analysis.

That's what I'd just like to leave you with.

And I took way too much time, but I appreciate your attention.

DR. KIRSTEN SMITH

"Okay... um... good morning everybody. It is an honor to be here and uh... hopefully the clicker will work for me. Okay — it's kind of working.

So... um... I just want to say that any views or opinions expressed today by me do not necessarily reflect those of Johns Hopkins School of Medicine, and these are my disclosures as well. I'm an Assistant Professor at Johns Hopkins.

So picking up where Chris left off — we have a lot of products all under the kratom umbrella, and again for all the reasons he just stated, it really is important from a clinical and, you know, forensic toxicology standpoint to really define what we mean by kratom, and I can't stress that we need a better taxonomy than we currently have.

So what I consider to be 'mainstream kratom' — like everything on here — I would draw a circle around this mainstream kratom. This has all been around awhile. The products that Chris just mentioned — they are not on here for a reason. I think it's just important to keep that in mind.

Oh... this clicker is horrible.

So... um... so... um... you know — in 2007 kratom first is documented in the case report literature. Dr. Boyer will probably talk more about that. And this really did come in a time in which we had a convergence of crises: psychiatric health, pain, chronic pain, and then of

course the polydrug epidemic. And we still have these epidemics, right? Has not changed. But kratom is part of this bigger story — in that people were looking for something.

And many of the... the reason I got into this personally is meeting people in treatment settings who were really not getting their needs met and they were looking for some alternative because the pharmaceutical industry and the medical industry was either not available or not helping them anymore. And so that's how I got into this.

And if we fast forward to... um... 2024, you know, we see a lot of the same. And I'm going to talk about how the motivations have changed, but who is actually using now.

Back to the prevalence estimates briefly — we really don't have great prevalence estimates. We need a better range than we currently have. Because I agree with Chris — I do think it's over 10 million. I don't know how many daily users there are, but it's certainly in the many millions. And for context, there are about 34 million cigarette smokers in America still. Right? So... so many millions — this is not niche as a marketplace. It is something that does require attention and regulation.

Most people are in their mid-30s on average. It is used between men and women pretty equally — unlike Southeast Asia where it's more males reporting use.

And actually the kratom-using population — with respect to income and education — looks a lot like the general population. There might be some more health conditions among this population, but in many other ways it's actually very... I mean compared to some populations I've studied that use illicit drugs, this population is very diverse and... you know... it doesn't look like just one thing, which I think is really interesting.

Most people are using daily at this point — or they're using recreationally and occasionally. We'll talk about that in a minute.

Those who are using daily are typically using two to four times per day on average.

And here is a bunch of stuff... right? It's a little overwhelming but I want to walk you through some of this because it's the most granular data we have to date with respect to kratom use patterns in the United States or globally.

What we did when I was still at NIDA is: I led a study — Chris was actually a mentor on this — and we ended up doing what's called ecological momentary assessment, where we gathered — across 357 participants — data at random and scheduled intervals on a study smartphone application.

So we know when people wake up, we know when they go to sleep — everything is very structured and standardized. We had over 13,000 unique kratom-use events that were identified during this 15-day period.

And we used those to do an analysis to see — you know — what are the patterns of use within this sample.

And you can see — hopefully see — we have Group A at the top; they're using most frequently, all the way down to Group E which is using the least frequently.

I want to draw your attention to the fact that — you can't really see it — but these are the hours of the day spread out at the bottom of that graph. And what this is showing us is that most people, no matter how much they're actually using, are tending to use more during the first half of their waking hours.

And this is important, because what we didn't see was binge-like use or heavy late-night use or things like that. We see people using this at times of day when they are actually trying to be most productive. And we know people in the sample were employed or in school. So these are people who are living their lives actively and using this as part of their daily regimen. And I think that's going to be increasingly important to keep in mind as we go.

This clicker is really just something else...

Okay, so this is a curve that is based on all of these thousands and thousands of data points, showing that across products — and I should say most products in the sample were whole-leaf products, not all of them — that the time-course of effects are peaking at about one hour. And we see this in the lab study; I have a clinical trial at Hopkins right now — we're seeing about the same thing.

So this is not a substance in which you take it and it's lasting for 12 hours. And it's not something you take and it lasts for 12 minutes. Right? It has a very nice, orderly time curve. And this is across multiple different doses — so it is kind of an average, it's not specific to a particular dose or product.

This here is showing motivations for use — and these are proximal motivations in real time. So when we look at why people are using — not broadly, but in the moment — we see that analgesia (pain-killing effects) is being sought out. But also — and this is really important — the stimulatory effects: increasing mood, focus, productivity, energy.

These are again what people would want as part of daily living. Right? They're not necessarily wanting to feel “high” or intoxicated — which I should say we did evaluate. We asked about that. And that did not even make it into this graph because it was not endorsed at a high enough rate.

People do endorse wanting to feel “good,” which is very different than wanting to feel intoxicated or high. And I think that is an important point to keep in mind.

So acute effects — very much in keeping with what people are seeking out: energized, focused, alert. We see some people not experiencing any effects, but that's actually because we had a long follow-up window, and so some of those acute effects had already peaked by the time we got the follow-up data.

But again, to emphasize: stimulatory and analgesic effects at low and at high serving sizes.

This is kind of an important finding because a lot of times people say lower doses are really for more stimulatory effects and higher doses are for pain-killing effects — and that is actually not what we're finding. Both qualitatively and when we look at things like this, it really is... I think when you get to a certain serving size, you do have a slightly different thing — but that's over 12 or 13 grams, which not everyone is using.

So instrumental use and compatibility with daily living — again this narrative of: how is this fitting into a person's life?

There's a lot of data I'm not showing — we know what people are doing when they're doing what activities they're engaged in — so there's a ton of stuff I'm not showing.

What this slide is showing is: what we ask people at the end of the day for these 15 days — which is to reflect upon how the kratom effects you experienced today fit within your daily roles and obligations. Were they helpful? Were they compatible? Were they a hindrance?

And, by and large — across all of our different groups — what we've seen here and what we've seen in another sample where we assess this is: people are overwhelmingly saying these kratom products are not only compatible with their daily living, but also very helpful in meeting their daily roles and obligations.

Which is part of this instrumental-use thing.

So, like, when I drink coffee this morning — that's instrumental use. And that first one — “compatible and helpful” — the way I like to help people understand this is: it is kind of like a cup of coffee. It is not only compatible, but it is helping me give this talk a little better than if I hadn't had my coffee.

That second one — “compatible but not necessarily helpful” — that’s more like a glass of wine, or a cocktail. I could have a drink at the end of the day; it’s not incompatible with my life. It’s not necessarily helpful either. So it’s not a hindrance.

And that’s the big takeaway: what we’ve seen in almost all the samples we’ve assessed this in is: it’s not a hindrance.

We will get into cases where it can be a hindrance, but by and large it’s not.

I want to touch on the harm-reduction point for just a minute, because this is — in my estimation — one reason why kratom was not scheduled in the first place. And I think remembering the public-benefit vs. harm of this is not a trivial point.

We know — and we’ve known for a while — even going back to Southeast Asia — people using kratom products in the United States to attenuate withdrawal from many substances.

And in this sample, people weren’t doing that because they had already gotten off those substances years ago.

We have people who’ve been using — in this sample — 93% have been using for more than a year; some had been using for over a decade.

And we see that Group A — who is taking kratom more frequently, not necessarily higher doses but more frequently — they were more likely than the other groups to be using as a long-term substitute for opioids. They were also more likely to perceive kratom as “lifesaving.” They were also much more regimented in their patterns of use — which is what we see with a maintenance-medication type regimen.

I’m not calling kratom a medicine — I’m making no medical claims here, and I’m talking about products, not pharmaceuticals. So this is very much real-world use that people are

adapting to their lifestyle and needs, whether it is for going and having a pre-workout drink — which we know is also happening — or for this harm-reduction approach.

And they're not mutually exclusive — many people are naming multiple reasons.

Physical dependence and addiction — this is... I am, believe it or not, the world expert on kratom-use disorder. Shockingly. There aren't many of us — Darin Singh in Malaysia would be my Asian counterpart.

And so what kratom-use disorder is, is essentially DSM-5 diagnostic criteria for substance-use disorder applied to kratom. And this is important because it is a very widely used assessment method.

So we — I and colleagues — have assessed this in this sample and in others.

What we see here is that oftentimes people don't meet criteria. And when they do meet criteria — you need two symptoms to meet at least “mild” — and we see that when people do meet a diagnostic threshold, it is typically mild to moderate.

And then when we look at the individual symptoms — which is absolutely very critical — we see that a lot of these symptoms are characteristic of what we call physical dependence — which is defined by tolerance or withdrawal. And that could be using to avoid withdrawal.

So again — when I drink my coffee this morning — if I didn't drink my coffee, I was going to get very fatigued, I was going to get a headache. That is a withdrawal symptom.

So we do know — I believe at this point it is fair to say — that daily use of kratom or regular use of kratom will lead to withdrawal or tolerance symptoms. But they are described as fairly manageable and mild or moderate in nature.

I'm actually doing a supervised-withdrawal study at Hopkins and we are looking at physiological and every other way of assessing withdrawal — that's what we're doing right now.

So going and picking up kind of where Chris mentioned — risky or safe or dangerous — we have very few published case reports on kratom-related morbidities and mortalities. It is a shockingly small number — and I have read every single one of them. Right? So we don't have a lot globally.

We have a few pharmacokinetic studies. In these studies — which I don't have time to get into — no serious adverse events occurred. We did have some GI upset — which, again, given the plant material, is almost understandable because it's kind of punishing.

I want to touch on the FDA single-ascending-dose study very quickly. I cannot read this far — and I didn't — so this setup... I'm going to wing it here, and we're going to see how well I do for those of you who can actually read this better.

But FDA did get up to 12 grams across eight cohorts — so this is 40 healthy adults, and within each cohort there were six who received kratom and two who received placebo.

Again — no serious adverse events. That's a big deal. That is part of how we define safety.

Vomiting was the only adverse event that occurred at a rate greater than placebo — and it was when healthy volunteers took 8 grams or more. And that's a lot of material to consume if you're not a daily kratom consumer. If you take 8 grams of anything, that's a lot of material.

And so the fact that vomiting was the only adverse effect that differed from placebo is really interesting — and it's also in keeping with what we saw in other studies where adverse events were observed — which were GI upset.

They did not see strong indicators of abuse potential. This is not an abuse-potential study — they are moving forward with that because they have found that this is safe and well tolerated.

And so I should also note that they did not have strong indicators within that abuse-potential paradigm in this drug-effects questionnaire — which is what they used — of feeling intoxicated or high. And that's in keeping with what at NIDA — when I was a postdoc there — we saw in a small substudy.

We had 10 kratom consumers come in; they took their own product and their own self-selected doses — between 1.1 and 10.9 grams — no adverse events.

And there's a lot of ecological validity here: the FDA study used well-characterized kratom material; these folks are taking whatever they're buying. So this is not selecting an ideal standard — this is seeing what is actually going on.

And, you know, unlike FDA, we did find some higher ratings of liking — which I think is interesting and actually not surprising — because if people are using kratom daily, obviously they like it.

So there's a question of: what abuse-potential indicators would be found among kratom-naïve consumers? And it might not be that high.

We did not see intoxication. We did not see impairment. That's both subjective report and objective psychomotor tasks.

In participant interviews, we did get some signals of serotonin excess — and that's important. So speciogynine and paynantheine — sorry, I'm not the chemist — but we know that there are some alkaloids that act on serotonin receptors.

And you know — a lot of people Google kratom and they get “opioid” that pops up at them — and that really, really irritates me. I do not like the internet when that happens because **kratom is not an opioid**. There’s opioid activity — but there is so much else going on with this botanical.

And I do think — well, not think; what we are following up on — is assessing the serotonergic-excess possibility at Hopkins now. It is something that we think everyone should be aware of. It’s not a danger at this point, but it’s something that needs further investigation.

And I should note that when people did describe having unpleasant events with kratom, none of them sought medical attention. So that’s probably a reason why we don’t have more published case reports — because a lot of people are not seeking medical care.

Last slide — this is a bunch of stuff shoved onto one slide, and it’s what everyone loves — they love the driving simulator.

We have a machine that is validated to detect impairment with alcohol and cannabis — and we had people, before and after they consumed their kratom, get into this driving simulator.

And we really did not see any indicator of impairment.

We had ONE participant who had a slight increase in standard deviation of lateral position and a lower coherence score — which is for that one specific task of many, many outcomes. That one participant did show a slight decline.

Everyone else — there was no trend in any direction. Some people improved. Some people stayed steady.

So again, we need to follow up on this — particularly with kratom-naïve consumers and higher-potency products — because a kratom-naïve consumer could have a different outcome.

I really hope this is the last slide — it is.

So takeaways:

Kratom — still a harm-reduction tool — but really increasingly used for recreation, fitness, quality of life, wanting to feel good

DR. EDWARD BOYER

“Thank you very much. My name is Edward Boyer. I want to thank you for the opportunity to speak.

A couple things to get out of the way: I’m an emergency physician. I’m a medical toxicologist. I don’t have any conflict of interest other than giving expert-witness testimony in kratom-related matters, even to the point that I haven’t accepted compensation from the American Kratom Association or any other trade organization.

And that kind of makes sense, if you think about it, because — you know — Chris has talked about the basic science of why these molecules work the way they do, and Kirsten has talked about why people use them in the way they do, and the effects that they have.

But I’m a medical toxicologist.

I deal with:

- poisonings
- overdoses
- adverse drug events
- drug interactions

And no pharmaceutical company in the world — no dietary supplement manufacturer, no food manufacturer — ever wants to test their substances for poisoning.

So whereas they have the great advantage of being scientists — I'm more like a... I'm slightly different. They can do controlled trials; the people who come to me have to have an event, and then somebody has to recognize that the event has occurred, and then make a connection with the substance, and then they will perhaps call me.

So a lot of the toxicity that is reported related to kratom is just purely by happenstance. And whenever you see happenstance you have to think in terms of biases that occur.

So I'm going to focus — you know — we've talked about the science, we've talked about the benefits, and I'm going to talk about the risks at this point.

When I think about the risks associated with kratom use, it falls down for me into two primary areas:

1. Seizures

2. Respiratory depression

So for seizures, to give you a little bit of context: This was many years ago, and Chris was involved in this case too.

It was somebody who was injecting an awful lot of Dilaudid. His wife told him to stop using drugs. He said, "Give me a couple of days." He learned about kratom — this was back in 2004 or so — so, you know, many years ago at this point.

He began using kratom three to four times a day. He did this for three years. And he noticed that, you know, he was more alert, he was paying more attention to his wife, he was getting work done.

Now you could say it's because he's not injecting full-bore opioids several times a day, or you could say it's related to the kratom — we can't say that one way or the other — but regardless, he notices that he's now paying attention to his wife.

And he decides to see if he can boost his level of alertness a little bit more.

So — being a junior pharmacologist — he takes a Methylphenidate tablet, because that's something you would use to increase your alertness. Didn't want to overdo it — he cut it in two. He took half the tablet and drank a cup of kratom tea the way he normally makes it, in the dose that he normally used.

And he suffered a generalized tonic-clonic seizure, which brought him to care at Massachusetts General Hospital, where they had never heard of kratom before.

So we got called — as toxicologists we went over — and after the evaluation we found that literally all he had taken was Methylphenidate. We were able to prove that what he had taken was kratom.

And, you know, we wrote this up as a case report — as a potential adverse event related to kratom — but you have to classify it exactly in that vein. It's a potential adverse event associated with kratom use. It does not prove causation. It does not prove that kratom caused the seizure.

But a couple things pop out that are really useful to keep in mind about seizures in the kratom world.

The first thing is:

Do people suffer seizures after using kratom?

Yeah, they do.

On the other hand:

Do people suffer seizures after doing literally almost anything?

Yeah, they do.

So we don't know how strong the evidence is just based on that.

So we have seizures associated with kratom — but now you have to look deeper at whether or not they've had a seizure disorder.

A lot of individuals who use kratom and have had seizures have a known seizure disorder — so it's not entirely surprising that they're going to seize again.

You want to look at the medications or other substances that people have used — and the majority, I'm not going to get into numbers, but the majority seem to have used multiple substances.

How those drugs interact is an area that must be investigated — if we had enough patients that we could actually figure out if there's a pharmacodynamic interaction between those two substances in that individual.

So if you look at the substances, you need to look at whether or not they are pro-convulsant.

Do the other things they've taken with kratom actually cause seizures themselves?

And in a decent proportion of individuals — certainly in Malaysia — yes, they do cause seizures.

So is it correct to say that kratom caused seizures in somebody who has a seizure disorder, or used another substance, or used another substance known to cause seizures with kratom?

Probably not correct — from a scientific perspective.

But what about those individuals who don't use another substance, who have no risk factors for seizures, who have no other substances they've used?

What about those folks?

Well, if you take those individuals — and this is a research study that has been performed with a small number of individuals — take those individuals with no risk factors, no history of seizures, no concomitant drug use, and do an EEG (electroencephalogram):

You put a whole bunch of leads on their head to determine if they have abnormal electrical activity in their brain that can predispose them to seizures.

The answer is: yes, they do.

So they did not seize up to that point — even though they're at increased risk of doing so — because their brain has that electrical activity at baseline.

So: Can we say that kratom causes seizures?

The answer is clearly no — you can't say there's causation.

Although there is an association.

Does somebody who has seizures following kratom use need an additional workup?

Yes.

We know from the cocaine experience:

There were an awful lot of people out there in the 1980s and 1990s using cocaine. There were some that would seize from time to time. But the number of people who used cocaine was vastly larger than the number of people who developed a seizure.

So a lot of toxicologists say that if you use cocaine and you have a seizure, you need to work up for a primary seizure disorder.

And I think that's probably a rational recommendation for kratom as well.

If you seize without any risk factors — without any concomitant drug use, without a pro-convulsant medication — you need to work up for a primary seizure disorder.

Now — I'm going to pivot a little bit now and talk about respiratory depression.

And what Chris didn't tell you — and what I don't think anybody has told you yet — is: I'm actually a chemist. I'm a synthetic organic chemist. And then I went off and did other things.

And a lot of my work was in the shape of molecules.

If I make a model of a classic opioid — like morphine, or heroin, or hydromorphone, or codeine — and make a molecule of it: it's a kind of T-shaped molecule. And that binds into the mu-opioid receptor.

Those molecules produce a lot of biological effects:

- euphoria
- analgesia
- pupillary constriction
- mood effects
- slowing of gastrointestinal transit time

So that T-shaped molecule goes into a receptor and causes a whole universe of biological outcomes.

Mitragynine is different.

Mitragynine is flat.

It cannot fold into a T-shaped conformation.

So it binds the mu-opioid receptor in a fundamentally different way.

Now — how can a T-shaped molecule have all these effects? Well, they're massive molecules. They flop around all the time. And if you bind in a particular way, then you stabilize the shape of the molecule that produces pupillary constriction and respiratory depression and analgesia and gastrointestinal slowing.

You stabilize the receptor that produces all of those things.

You take a flat molecule and put it in there — and it does not do all of them. It only does some of them. And even the ones that it does, it doesn't do as well as the native molecule.

So the classic mu-opioid receptor ligands — the morphines, the heroins, the hydromorphones, the oxycodones — they have that T-shaped structure, and they produce the whole universe of adverse events that we expect from opioids.

They are called unbiased ligands — they have no bias toward one particular biological activity or another.

Mitragynine — being a flat molecule — stabilizes only a couple of receptor conformations, and not very well.

It is considered — and called in the literature — a biased mu-opioid receptor ligand, because it doesn't cause some of the things the classical opioids do.

It does not seem to produce respiratory depression to the same degree.

Now — can I knock that out? Perhaps I can.

Perhaps if I give enough, I can stabilize all the conformations and produce the universe of biological activity. But I don't know the dose at which I could do that. I don't know the other drugs that I can take that will do that.

So to say that kratom and mitragynine induce respiratory depression — I don't know that there's a lot of strong evidence for that.

In fact, there's kind of some evidence in the opposite direction.

We know — back from the 1960s — that if you took a rat and drew up a whole bunch of mitragynine in a solution and injected 600 mg/kg — they would have no respiratory depression whatsoever.

So there's animal testing that suggests strongly that there's no respiratory depression.

There are also pharmaceutical products — not in common use, but patented, with IP protection — that are FDA-approved that are biased ligands. So big pharma recognizes that biased ligands are a potential way to go in creating new molecules.

So — that's really where, for me, the rubber hits the road with kratom use.

And I look forward, when all are done, to take any questions related to it.

Thank you very much for your time, and again for the opportunity to address you."

DR. JACK HENNINGFIELD

“Thank you. Can you folks hear me okay?

Great. It’s an honor and a pleasure to be here.

Every time I have an opportunity to participate in an event in the Capitol, I take a walk around the Capitol when I’m done and just marvel at the fact that we can actually do this — and have input — and interact with people like Representative Bergman and Pocan who help make this happen.

Dr. Volkow isn’t here, but you’ve heard a lot of the research that is being done because Congress funded NIDA to do this research. It comes together.

It’s great to be the last one, because part of my mantra is: How do we get FDA in the game?

You heard from Drs. McCurdy and Smith — the diversity of products in the marketplace. And, well, what do you do about that? FDA is the best organization in the world, hands down, to figure out:

- what should the standards be
- what should the labeling be

Over-labeling is as bad as under-labeling.

Should products say “Caution: causes seizures”?

I’ll bet FDA would conclude: No — you don’t put a label on just because something might do something that’s dangerous anyway.

That being said — next. Let me see. Hopefully this will work — if it doesn't, I'm going to ask you folks to do it.

Disclosures:

I consult in this area.

I've been working in addiction research since 1971 in neuropharmacology, and working in drug regulation since — that was part of my job in 1980 at NIDA — working with FDA and DEA. And I still work with FDA and DEA. And I respect what they do. We don't always agree, but they're all trying to do what we're trying to do — get it right.

Just a couple things I'm going to mention — a little bit of the history you've already heard. A lot of the history that FDA ignored.

When people ask me:

'How did FDA get it wrong?'

My answer is:

Well — they came to the table late, around 2011–2012.

And they didn't consult with NIDA researchers.

And they didn't consult with Southeast Asian researchers who had been studying it for decades.

There are a lot of promising developments — and that includes the findings now. The science base is incredible. You just heard from some of the pioneers in science in this area in the United States. And they stand on other pioneers.

Dr. Smith — it's amazing — she's become a pioneer in just a few years. She's a nova pioneer.

And Congressional encouragement of FDA is happening — and that's encouraging — because a lot of times FDA needs encouragement from the United States Congress. They need Congress to say:

“Why don't you do this?”

“Give us cover.”

“Help this move forward.”

Let me go back with just a couple of things about some of the history.

Guess what?

In 1972, Smith, Kline & French allowed their scientists to publish the results of their exploration of mitragynine as a new pain drug.

And you already heard a lot of what they found back in the late 1960s, 1970s. I think they were allowed to publish it because — for whatever reason, maybe patent reasons — the company decided not to go forward. But you can see what their conclusions were:

“In our search for a potent yet safe analgesic possessing a profile of qualities that are superior to existing agents, mitragynine looks like it's the real deal. It produces some pain effects, low respiratory depression, safe profile. **It's not an opioid.**”

Somehow FDA missed that — when they said it's a deadly, addictive, narcotic-like opioid.

As a kind of epilogue — how would the world have been different in pain management if they had gone forward with it? But they didn't.

By the way — that's their summary. Anybody who wants my slides — they're publicly available and I'll send them to you.

Now — as Congressman Bergman stated earlier — with FDA:

“We've got to work with FDA.”

So about four or five years ago, Mac Haddow said:

“What study would make a difference?”

I said:

“Even though we know kratom doesn't cause respiratory depression like an opioid, we need to do FDA's test.”

Can you get the funding?

And some of you in the room may have contributed the funding — I don't know who.

I said:

“The deal is: we’re going to show the data to FDA and NIDA before you see it. And we’re going to publish it no matter what the results are.”

We did that.

And the curve with the blue is flat as a pancake — that’s respiratory effects under a huge range of doses.

And the green is the oxycodone control, and that produced the Death Valley effect on respiration — and deaths — in some of the animals.

Next slide, please.

This was a historic study that both Chris and Ed already mentioned. You know — science — a lot of it is we just, by accident and fortuitous events.

So Dr. Boyer has somebody coming saying:

“Hey, what’s going on with this guy?”

And Dr. Boyer calls Chris McCurdy.

So here you had a lead kratom researcher and a lead toxicologist — an opioid guy who really knows pain and opioids — looking at this.

And look at their conclusion:

“We report self-treatment of chronic pain and opioid withdrawal with kratom. The predominant alkaloid, mitragynine, binds mu-opioid receptors but has additional receptor affinities. It might augment its effectiveness.”

FDA got it wrong.

It wasn't just a morphine opioid-like receptor.

It's a diverse substance.

And that probably accounts for a lot of the benefits.

It's not a powerful analgesic — but it does have analgesic effects.

But it does a lot of other things that are important.

Science. I love it.

Next slide, please.

In 2018, the FDA was stopped in its tracks by science.

Now — that wouldn't have happened without advocacy and consumers — but it was science.

After the FDA bombshell, with the Commissioner himself saying:

“This is a deadly, addictive, narcotic-like opioid.”

What did NIDA do?

They ramped up research.

NIDA and DEA were actually closer aligned than FDA at that time.

We see Dr. McCurdy and others were already researching — but a lot more money came in. And I would say that is a success story in funding NIH and giving researchers — who are creative and dedicated, like Dr. McCurdy and others — the resources to run with it and give us what we need for regulation.

We don't know everything — but anybody who says:

“We don't know enough to start regulating”

doesn't know anything about regulation.

There's plenty for FDA to start with. Whether you're regulating water, food, drugs — you start with what you've got. And your regulations evolve as you learn more.

We've got a lot of information.

Scientists' letter to the White House — we wondered whether anybody would read that — and it made the case that:

If you ban kratom, people are going to die, because people's lives depend on it.

Secretary Azar turned to his Chief of Staff and said:

“FDA is probably right, but these seem to be credible scientists — let’s look at it.”

What did he do?

After their investigation, they found the same things.

They investigated independently — and actually, later, Dr. Azar, in a tweet to Dr. Gottlieb said:

“FDA’s recommendation was rejected because of embarrassingly poor evidence and data, and a failure to consider overall public health.”

“If kratom is fueling the opioid epidemic — prove it.”

I knew it wasn’t.

Then in June, NIDA and the College on Problems of Drug Dependence — all agreed:

“This is an important area in public health and science.”

In that single meeting, Dr. McCurdy had a really historic symposium on kratom — because he brought together a lot of the lead researchers. Five FDA people were there — I know because they were sitting right behind me.

Dr. Smith gave — and she was a baby at the time, 13 years old — she gave one of the Hot Topics presentations, which are very well attended, on people with heroin addiction using kratom to stay off heroin.

NIDA brought in Malaysian experts and had an international kratom panel.

Here's where people were saying:

“Kratom is an instrumental substance.”

And all of these sessions were attended by FDA, NIDA, and DEA.

That was a very important point in stopping FDA's progress.

Now — I'm going to bring a little pain to the audience...

You've heard a lot about pain. Two weeks ago — I believe the first in history — a kratom presentation at the annual PainWeek convention occurred.

Dr. Rafa and I co-chaired it.

When the audience was there, we had one of the larger audiences, and I asked:

“How many of you are treating patients with kratom — or your patients are using kratom?”

More than one-third of the hands went up.

So why are you here?

“We want to learn more.”

They want information from FDA — they’re not getting it.

But why is kratom so helpful in pain?

And again — there’s no pain indication. Kratom is not approved for pain or any other indication — that’s important to recognize.

Next slide.

However — self-reported pain is a major reason people report using it. And what’s really interesting is that people report:

“I got my life back.”

“It made things better.”

What’s going on? Especially since it’s not a powerful analgesic — but it does meet analgesic tests in drug company studies and others.

Well — it has effects that go along with chronic pain.

Next slide.

And this is from a paper I presented at a psychedelic conference. We're learning that psychedelics like MDMA help a lot of people with chronic pain. What's going on there?

They do not have analgesic effects.

Well — chronic pain is a multimodal problem. It includes psychological components. And multimodal treatment works better.

And the pioneers in a lot of this were the U.S. military in the 1990s — discovering that soldiers with blown-off limbs did better if you gave them counseling — divorce counseling — what's that got to do with pain? Psychological counseling.

But the people who got counseling, multimodal treatment — maybe antidepressants — they did better with pain recovery, getting their lives back, and they needed less opioids, and got off opioids sooner.

Well, as you already heard from Drs. Boyer and McCurdy and Smith — kratom isn't just an opioid. It has some opioid effects — but it has emotional effects, mood-altering effects that are beneficial. A lot of those are why people use it.

When I was first exploring kratom more than 10 years ago, I thought:

“What's going on? People are using it for fibromyalgia? For post-traumatic stress?”

Well — fibromyalgia is pain, sleep, and depression.

Kratom affects all of those.

Next slide, please.

So — the Lancet. There's a revolution in pain treatment right now — and the revolution is a reset. We have to treat chronic pain focusing more on the emotional and physical problems, helping people get their lives back — not giving up on relieving pain — but doing all of this.

And kratom looks like, in one plant, you get a lot of these effects.

Plus a bonus.

What's the bonus?

What pain medicine increases alertness, mental acuity, and helps you get your job done better?

Next slide, please.

Okay — has anyone heard that we might have the first downturn in drug-overdose deaths?

A really hot topic.

Nora Volkow weighed in on it in the last couple days. Everybody's saying:

“What's going on?”

Well — is it real? Yes — but it's complicated.

And guess what — it's not happening with stimulants and cocaine. They're flat or going up a little bit.

But opioid deaths are going down.

That was real.

Will it continue? Nora Volkow says:

“We need more harm reduction.”

“We need more naloxone.”

She didn't mention kratom — but I think this makes it more important than ever.

We know there are some people whose lives — dangle by the thread of kratom. We don't know how many. And kratom hasn't been mentioned in these discussions, but more than ever we need FDA to step in and regulate — and not demonize it.

Don't scare people away from what could be life-saving.

And why is that so important?

It's always important — but whenever there is an addiction going on, it's really important.

You've heard the old mantra in addiction treatment:

“Denial — it's not a river in Africa.”

It's part of the addiction process. People often love to hear that the treatment is bad — so they can stay on what they're taking.

We need honest, accurate information out there — and kratom, we think, is part of that solution.

How big a part? FDA — we have to get you at the table helping.

Next slide.

I showed this slide — some of you may have seen it. I presented it at an FDA hearing in 2018 on new treatments for opioid dependence, because that's when FDA was saying:

“It's not approved — so people should stop using it.”

And I presented this slide — and I heard later that some colleagues liked it, and some of the people at FDA did not like it.

I said:

“You don’t take the life preserver away from somebody struggling in the ocean because it’s not Coast Guard approved.”

Okay — last slide.

Sustaining and accelerating the decline of overdose deaths.

How many of you recognize that guy before you saw it?

I worked for him. I was assigned to work for him in 1981 until he died. He was a harm-reduction guy.

Dr. Koop understood that much of public health is harm reduction — clean needles, less harmful substances.

He said — about nicotine:

“If you’ve got to be hooked on nicotine, be hooked on a product that’s less likely to kill you.”

This is very much what public health is about.

But we need — and kratom also has the potential to help with alcohol and stimulant-use disorders. That’s part of NIDA’s interest in it.

But what do we need?

Might need more Congressional support.

Ensure that kratom remains accessible nationally — the Kratom Consumer Protection Act. More states — ideally we need it federally. Everybody should be protected — not just some people.

FDA and others criticize kratom for not going through FDA or having a New Dietary Ingredient notification — but FDA is denying all of those.

And guess what — that process, if they do it, they learn a lot. Because for that product, they have to decide:

- what are the standards
- what are acceptable ratios of substances — which Dr. McCurdy told you are evolving
- what is an acceptable kratom product

Then the industry knows what they can market.

And they can make sure the legitimate players meet that profile.

And if you don't meet that profile — you're not going to be in the market.

Industry can't do that.

FDA has to do that.

So there are a lot of things that we need Congress to maybe nudge — encourage FDA.

Which, you know better than anyone — Representative — that letters, hearings, meetings...

Recently, Representative Bergman and Correa — who head the bipartisan psychedelic caucus — issued a Request for Information to help FDA move forward more quickly on psychedelic medicines development. They did that after the disastrous hearing with MDMA in June.

I think that will get approved, but it's going to take longer.

And they're helping move the process along.

So — last slide.

Conclusions and pleas, and I think most of this has been covered:

- With support of research to NIH — this is an amazing success story. In the last 10 years the research publications have skyrocketed.
- When I watch Chris present — what we know now compared to five years ago... Five years ago he said, "Major constituent of kratom is 7-hydroxy." Now we know: No — it's a metabolite. It's in very low levels. If it's in high levels — it's probably been spiked.
- We know a heck of a lot more.
- Although the overdose epidemic might be subsiding — we hope it'll continue — we need kratom in the game and legitimized, not demonized.
- Don't demonize what can be helping.
- NIDA has walked that path carefully — they say kratom is not approved as safe and effective — but a lot of people are using it. They do not say "don't use it."
- DEA — that's part of the reason DEA opposed scheduling and walked away from trying to schedule it.
- Thank goodness for Dr. Azar for stepping in.

Kratom is an under-employed health asset.

How do we leverage it more?

Balanced FDA oversight can protect and guide those for whom kratom is their lifeline — and provide standards for the industry. Legitimate industry — anywhere — wants to know what the rules are and what the label should be.

FDA can do that better than anyone.

Also — I would love to see what's happening in the psychedelic area from Congress — the Department of Defense and Veterans Administration have gotten in the game.

In the last budget — the Defense Authorization Bill said:

MODERATOR

“Thank you — thank you to our panel of scientists.

We'd like to move quickly to the next panel, which is a group of kratom advocates, and in the interest of time — we only have this room until 11:30 — I'd like to ask the panelists, the next panelists, to limit their comments to seven or eight minutes maximum.

And thank you again to the scientific panel.

I would ask the next panelists to grab your little placard and bring it up with you. Thank you for doing that.

We have four panelists. I'll introduce them first, and give you their background.

We have first: Peter and Kelly Guidry, founders of Forgotten Not Gone, a remarkable veterans organization dedicated to stomping out veteran suicide.

Michael Krawitz — Director of Veterans for Medical Cannabis Access.

Melody Wolf — a kratom consumer from Michigan.

And Heidi Sakur, a kratom user from Wisconsin.

Their personal stories and advocacy efforts reflect the very real, positive impact kratom has had on their lives and countless others.

And without further ado, I'm going to turn to Peter and Kelly. And please remember to push the button in front of you — yeah, push it and it'll go red — and please, let's limit comments to about seven or eight minutes so everyone has a chance to talk. Thank you."

—

PETER & KELLY GUIDRY

(Kelly speaks first)

KELLY GUIDRY (Forgotten Not Gone)

"Thank you so much. My name is Kelly Guidry. I am the founder — well, co-founder along with my husband Peter Guidry — of Forgotten Not Gone. I am also an Air Force veteran.

Our nonprofit is suicide prevention through veteran peer-support outdoor recreation.

Um... I would just like to thank you so much — thank you to Mac... thank you also to Kelly Dunn as well... and to Jeremy Steing — they have supported us throughout our career of doing this.

The biggest thing that I really wanted to point out was a couple of things.

The way we were introduced to kratom was through a supplement store — so I've never experienced the gas stations or anything like that. It was always introduced to me as a supplement.

My personal story is — I was addicted to pain pills and alcohol through the VA. They gave me hydrocodone, which is an opioid, for my fibroids. I was in severe pain all the time. And that is a huge issue when you want to come out of addiction — when you want to get into recovery — the biggest thing that scared me the most was chronic pain.

I don't think people really understand what it's like to live second-to-second in pain.

So I really wanted to stop using the hydrocodone. But I knew that I would be in pain — so it was very frightening for me.

My husband actually started using kratom before I did because I was, like I said, severely scared of everything.

I just celebrated 13 years of sobriety — September 17th — so it was really important to me to still find something to help me cope with the pain but also have a quality of life.

And I would have to say that kratom did exactly that for me. It helped me have a quality of life.

And these last 11 years of doing our nonprofit — I would not be able to get up every morning and help my veterans, facilitate trauma healing, if I did not have it.

First getting over the pain, and then also being able to educate them and share something with them to help them get over it.

Just really quick — in 2021, women veteran suicide did go up — also civilian as well. I feel like we're at the precipice of something that's huge. If we don't do something, it's going to get really, really bad.

Women typically do not kill themselves — unalive themselves — and they are using firearms, and it is at a high, high rate.

And if we don't step in right now, something is going to be a lot, lot worse.

So I just want to thank you so much again.

Kratom has given me my life back.

I am not a daily user — I use it when I need it.

But having that resource to go to helps me have a quality of life, and I don't know where I would be without it.”

—

PETER GUIDRY (Forgotten Not Gone)

“And I’m Peter Guidry, the other half of Forgotten Not Gone — co-founder.

It’s been an amazing journey the last 11 years.

Having been medically retired from Nellis Air Force Base back in 2002 — you know — leaving the military being sick at that time, it led to a lot of severe depression.

And it was so bad, the first 10 years, that — you know — I came to a point that I wanted to take my life.

And I wanted to read a statistic real quick — it says that there’s actually 6,300 suicides among veterans every single year.

When we first started 12 years ago, it was 22 veterans dying a day — that was the number. But through the advocacy efforts, that number went down — we’re seeing maybe about 17 veteran suicides at that point.

But the reason veterans are dying is because they’re suffering in silence at home.

A lot of us are physically sick — we’re embarrassed because we used to be functioning at a high level, and now because of what happened to us in the military physically — you know — our bodies are broken. So it leads to a broken spirit.

So when I was seeking help at the VA, the medicine that they gave me for my back spasms caused more nausea, vomiting, vomiting blood. I started getting sicker and sicker because of the medication that was FDA-approved that the VA was giving me.

And it wasn’t until I was able to find kratom at a local supplement store called Pure Health in Las Vegas that I was able to get a different solution to overcome the physical and the chronic pain that I was dealing with.

So on every time that I ride — we usually do about three outreaches a week that we're doing right now — and we log 22 miles with our recumbent trikes.

And we have anywhere from 10 to 25 veterans out with us — they're from all walks of life, all ages — some 70-year-young Vietnam guys with us.

But a lot of those veterans now are finding the ability to participate in the veteran peer-support outdoor recreation we offer because of the aid of taking kratom. It helps them control the pain a little bit, the inflammation, to be able to get out of the house and work on the mind-body connection.

But before I had this as an aid in my personal toolkit — again, the medicine that was FDA-approved that the VA gave me was causing so many side effects that I was vomiting blood and I was very sick.

And 12 years ago, up to that point, I had been walking on a four-prong cane.

In the last two years I am no longer walking with a cane, but I'm still having the same physical disabilities, the same pain, the same limitations that I had 20 years ago.

But again — my personal kratom story is: it has allowed me to participate at least three times a week for the last 10 years in veteran peer-support outdoor recreation.

And in the last year, I've been on over 100 outreaches with different veterans that were suicidal, that were coming from different homelessness, battling different doom-and-gloom stories.

I was able to participate with hundreds of veterans in the last 12 months — all because I was able to control my pain because I had a new tool, a new solution, over suffering with the kratom.

So — I thank you guys for allowing us to be able to tell our personal story of how it helped me as a disabled veteran, not just have to give a doom-and-gloom story of what's wrong with me — but I've been empowered to be able to overcome my disabilities and have a new purpose in life again.

And I'm just filled with joy because it's a dream come true to finally overcome what happened to me 20 years ago in the military — to do something worthwhile in my life.

So — thank you again for your time.”

MICHAEL KRAWITZ

“Yes — uh, thank you.

My name is Michael Krawitz. I am a disabled United States Air Force veteran. I'm a non-paid advocate — I'm a volunteer.

I just wanted to read a quick, prepared statement — hopefully it's a shorter statement — and open up to questions.

As an advocate, I did work inside the VA system for about 10 years, leading up to the point in pain management where we were able to pass a national Veterans Affairs policy on cannabis back in 2010. So I... I do have some experience working as an advocate with individual vets and inside the system.

Pardon my voice.

Thank you for inviting me to discuss a pressing issue affecting the health and well-being of millions of our nation's disabled military veterans.

As they return from service, many of these brave men and women face a multitude of challenges, including:

- physical injuries
- psychological trauma
- chronic pain

And while our veterans deserve the utmost care and support, it is imperative that we explore innovative and compassionate approaches to address their needs.

One such approach lies in the growing body of evidence supporting the therapeutic potential of plant-based medicines such as kratom and cannabis.

These natural substances have been used for centuries to alleviate pain, reduce anxiety, and improve overall quality of life for patients suffering from chronic conditions.

These plant-based alternatives may offer a much-needed respite from the conventional pharmaceutical treatments that often come with significant side effects.

The vast majority of disabled veterans are looking to spend more quality time with their family, doing the things that they love to do. They don't want to be heavily medicated.

Many veterans have said to me that the VA hospital sent them home with big bags of pills and told them, "You know — go home and feel better."

Kratom, a tropical tree native to Southeast Asia, has gained attention for its potential to alleviate pain, reduce fatigue, and improve mood.

Studies have shown that kratom may be particularly effective for patients suffering from opioid addiction and chronic pain, by providing a natural alternative to highly addictive opioids. Kratom might help reduce the prevalence of substance abuse among our veterans.

Cannabis, another plant with a long history of medicinal use, has also demonstrated promise in treating a variety of conditions — including chronic pain, anxiety, and post-traumatic stress.

For patients struggling with these debilitating conditions, cannabis may offer a safe and effective means of treatment that promises to lower suicide rates and more effectively treat conditions like nerve-damage pain, pain that is often difficult to treat with opioids alone, even at relatively high doses.

The synergistic effects make cannabis an adjunct treatment, and it explains why many patients can dramatically reduce their pill count of opioids when successfully employing this plant-based medicine.

The growing body of evidence suggests that these substances may offer a valuable option for patients seeking relief from their symptoms.

Variability in plant-based preparations presents a big challenge to conventional medicine delivery systems that require they live up to their stated intention to deliver individualized care.

The voluntary veteran group that I am part of — Veterans Action Council — is working on strategies to affect federal patient access to medicinal cannabis products. These products are now exclusively available under specialized state medicinal-access laws.

Veterans Action Council hopes that with the imminent rescheduling of cannabis under the Controlled Substances Act, we'll be able to create a national access program that doesn't leave anyone behind.

The medical-care industry has a critical role to play in ensuring that veterans have access to safe and effective treatments by integrating plant-based medicines into evidence-based treatment plans.

Medical systems can help improve the quality of life for millions of Americans.

However — it is essential that policies enable a compassionate and non-punitive approach to these substances.

One significant barrier to the use of plant-based medicines is the fear of retribution, such as urine drug-testing. Such policies can deter patients from even seeking help and may even exacerbate their symptoms.

To address this issue, all medical professionals must implement policies that prioritize patient care over punishment.

It is important to keep in mind that it is never ethical for a medical professional to withhold a treatment as a form of retribution or punishment.

So, in conclusion — disabled United States military veterans and all medical patients deserve access to safe and effective treatments that can help them live meaningful and fulfilling lives.

Plant-based medicines offer a promising alternative for those suffering from chronic pain, anxiety, and other conditions.

By integrating these substances into evidence-based treatment plans and adopting a compassionate and non-punitive approach, we can help ensure that our nation's veterans receive the care they deserve.

And by integrating plant-based medicines into our medical treatment plans — we can help keep more of our patients in care.

A policy that may help to create more success stories rather than just feeding the drug war more statistics.

Thank you.”

MELODY WOLF

“Thank you.

My name is Melody Wolf, and I am from Kalamazoo, Michigan. I am a 10-year daily kratom consumer, and I'm here to talk to you — to try to convince Congress to keep it legal and to reasonably regulate it.

I am one of the people that Dr. Smith talked about — I have fibromyalgia. I also have spinal stenosis, sciatica, rotator-cuff tendinopathy, arthritis, and probably there will be a few more things as I get older — I'm sure that will come along.

And I just wanted to come here today and talk to you about how I just feel it's extremely important — not just for my life but for all of your lives — because September is Pain Awareness Month as well as Suicide Prevention Month.

There are over 50 million chronic-pain patients, and some people say up to 100 million chronic-pain patients. So that's a pretty significant portion of the population.

And I was reminded of a quote this morning — it said:

“Best-laid plans of mice and men often go awry.”

Well — we all have our best-laid plans. But how often do we end up at our destination?

I was reminded when I was in here this morning that I had dreams of becoming a Congressional page when I was in high school and I wanted to come to these halls of Congress and hearing rooms — and I thought, here I am today.

So that is an interesting way that I got here — was through kratom.

My pain started when I was about 40 years old, and it just started where it hurt to hold my husband’s hand. And then it continued to get worse and worse and worse, and the conditions continued to lay one on top of another, to the point where I was bedridden, needing a walker, and even a wheelchair at times.

And I just never thought that this would happen to me. And I don’t think most people do think that it’s going to happen to them.

Because like I was talking about — having the best-laid plans — I had really good plans. I got my bachelor’s degree in international business, I married my childhood sweetheart, I had three children, we were involved in our community, involved in our church.

And then the pain started — as I said, first of all holding my husband’s hand. And that got so bad to where they sent me to the Cleveland Clinic, and I’ve also been to the University of Michigan Rheumatology as well, pain clinics and many, many specialists.

At one time, I was put on eleven medications, one of them being the strongest prescription dose of the fentanyl patch.

And that is when I think our family started a downward spiral.

It made me mean and angry. I was in so much pain anyhow — it helped me tolerate the pain, but for me it really didn't take it away. So for us, life was pretty dismal.

My kids were like 6, 8, and 10 at the time. And they had to have fans on in their bedrooms at night because I would be laying on the couch crying in pain every night.

And there is no way for a kid that they have to grow up like that.

And the effects on our marriage were not... they were probably about just as bad as the effects on our children.

We all know how we would hope to react if a spouse has a chronic condition such as chronic pain — but my husband did not react very well. And then because of me taking fentanyl I didn't react well to his reaction. And it just continued to spiral.

But because of kratom — on August 10th we renewed our marriage vows with our family, with all of our family and friends present. And it was just so nice to have our kids — who went through all of this with us — be a part of that.

And kratom continues to have far-reaching effects for my family.

In 2022, my daughter lost her child care and it was two weeks before she went on maternity leave and she and her husband were like, "What are we going to do?"

Well — I got a phone call saying, “Mom... yeah... would you like to move in with us for about 9 or 10 months?”

I tell you — I did not even have to ask my husband. I said, “Don’t you have to talk to Dad?”

“No — I’ll be there.”

So... it was just shortly after that that I moved in with them and I was able to take care of my new granddaughter.

And they live about five minutes from the Detroit Zoo, and so I took her that summer about a dozen times to the zoo. And I was able to push her stroller around in the same zoo that my children had to push me around in a wheelchair.

And then I was just thinking about all the economic situations now and how high childcare is — they would have had to pay \$50,000 for that year for a full-time nanny. And as most young couples do, they would have made it work, but it would have decimated their income and made it hard for them to go on — you know — families want to build a savings account and get a house.

But anyhow — it’s my pleasure now to be there for my adult children and to call them my friends. And through the past years it’s been my pleasure to hear them say:

“Mom, I’m proud of you.”

And I just got a message from my daughter before I came up here saying:

“You’ve got this, Mimi.”

(That’s what my granddaughter calls me.)

And so I was not a mom that they could be proud of — and so now I'm just so thankful I have the chance that I can be.

Remember when I said that there are 50 million to 100 million chronic-pain patients? That means there's a one-in-six chance that everyone in this room will become a chronic-pain patient.

Certainly you're going to know someone that has chronic pain.

And I want that tool in the toolbox.

We need every tool in the toolbox.

And I need kratom to be one of them — because should you or a family member become a chronic-pain patient, bedridden, using a walker or a wheelchair:

I want kratom to be available for you.

I want it to be accessible.

I don't want to ever lose my mobility again — and that's what losing kratom would mean to me.

And I just want it available for all of you and your family members.

So let's don't... let's make sure that we don't plan to fail. I want us to get our plans in order and make sure that we keep kratom accessible and legal for all of us.

Thank you.”

HEIDI SAKUR

“Thank you.

My name is Heidi Sakur. I am a nurse practitioner and healthcare executive who retired early due to disability.

Despite struggling with many health issues in my life, I pushed through work and school — finally earning my Doctorate in Nursing in 2011.

Unfortunately, by 2015 my health issues led to severe pain, an inability to stand longer than a few minutes, fatigue, frequent choking, and gait instability.

Finally, after extensive research and many specialist visits — I was diagnosed with several rare congenital disorders. These include:

- Chiari malformation — that’s when the cerebellum or the lower part of your brain is beneath the opening of your skull and it was putting a lot of pressure on my spine, spinal cord, and on my brainstem; it actually flattened part of my brainstem.
- Tethered spinal cord — and that causes severe pain in my trunk and my legs.
- Lipedema — causing swelling in my legs and my arms.
- Ehlers–Danlos (hypermobility type) — a connective-tissue disorder that makes my joints very unstable and leads to chronic pain.
- Hereditary alpha tryptasemia — that makes me very drug-sensitive.
- Severe osteoporosis with multiple spinal compression fractures.

So — for years I was unable to find a tolerable and effective way to control my pain and fatigue.

Even after brain surgery, I could not tolerate the oxycodone — I only took it for a few days. Unfortunately, it made me very dizzy and unstable and it was a real risk for falls.

I am so drug-sensitive that even Tylenol makes me sleepy — so I can only take that at bedtime.

And of course — the NSAIDs led to GI problems.

Luckily — my son introduced me to kratom.

I like to say:

“I gave him his life, but he gave me mine back.”

Kratom has been a miracle, reducing my pain enough to exercise and stretch — which prevents further debility and injuries from my chronic conditions.

Most importantly — it has allowed me to enjoy our three grandchildren.

Like Melody said — being a grandmother is the most precious thing in the world. And not being able to play with them or participate in activities would be devastating.

Kratom has given me hope for the future.

I am no longer a spectator in life — but a participant.

It does not impair my thinking — and I have never had side effects like with other pharmaceuticals.

Regrettably — I only experience this comfort in Wisconsin and in Illinois — because kratom is banned in Wisconsin, where I have lived for 31 years.

Due to the prohibition of kratom in Wisconsin, I divide my time between Wisconsin and Illinois weekly. In Illinois, kratom is legal and provides me with the essential relief from the constant pain I endure.

The necessity to travel imposes a significant physical, emotional, and financial burden, as I must maintain residences in both states.

My husband's long-standing career in real estate and public service prevent us from moving to Illinois. Yet the legal constraints on kratom mean we are too often separated, and I really miss him.

This is also a hardship for my 91-year-old mother, who lives with me and endures the weekly travel between states.

I fear that we will not be able to continue this forever.

The burden is not just mine — it's shared with my loved ones as well.

God made the natural kratom plant perfect.

I ask you to keep it safe and accessible by passing the Kratom Consumer Protection Act.

And I thank you so much for being here.”

MODERATOR / CHAIR (American Kratom Association)

“One of the important takeaways I got from Heidi and Melody is how great it is to be a grandmother.

It ain’t so bad being a grandfather either. In fact, I like to refer to Wayne Gretzky’s quote all the time, where he said:

I really appreciate the panelists — the first panel that talked about the scientific aspects of kratom,

and this panel, which talks about the more personal side of how it’s impacted their lives or lives of friends or people that you’ve advocated for.

As a former policymaker — I like to call myself a “recovering politician” — as a former policymaker, it is incredibly important to get the facts.

You know, during COVID we heard over and over and over:

Well — that’s all we’re asking here today:

Follow the science.

In fact, the American Kratom Association was created really with one overarching goal — and that's the most important thing to us — and that is:

And to stand up for the consumers — that's really what we're all about.

And in my own state of Arizona, our legislature has passed the Kratom Consumer Protection Act. Several states in the country have done so.

But we really need a federal act.

There needs to be consistent policy across the country so you don't have to switch — move between Illinois and Wisconsin.

And really all this is about is protecting consumers.

We don't like the bad actors that adulterate products. In fact, our association has reported many of those bad actors — who either adulterated products or made health claims that they're not authorized to make — and we've given those to the FDA. They haven't done anything with it.

And so our goal out of all of this is to make sure that kratom continues to be safe and legal, but also that the industry is policed, and we make sure that bad actors are routed out of existence, and that consumers out there have access to clean, unadulterated products.

And so, for those of you that have listened to these panels today, we appreciate you being here.

Please — those of you that are staffers on the Hill — please go back to your bosses and let them know to talk to the cosponsors of this bill. There's a companion bill in the Senate as well.

And please ask them to co-sign.

We need this bill to be heard and we need it to be voted on.

And we need good, sound scientific policy coming out of the U.S. Congress.

And that's really all we're asking.

So, that concludes our panels for today.

And for the first panel and this panel — could we just get a group picture?

Yeah — we can do a group hug too if you want — but a group picture, right up here in front of the stage over here.

So — thank you.

And we appreciate your support.”

[Music]

END OF BRIEFING